

Important drivers of AI on edge

1. **Affordable cost.** The product and the technology has to be suitable at a reasonable price-point, both for general consumers and enterprises
 2. **Strong use-cases.** Relates to devices having intelligence rather than just control features. Examples include smart wearables and Alexa
 3. **Mobility.** Ease of carrying around a device such as Alexa Echo Dot, which can be kept in a pocket
 4. **Speed.** Reduced latency and fast response time
 5. **Security of data.** It is the biggest concern. So people prefer storing data on edge rather than on cloud
 6. **Reduced downtime.** More downtime = More financial losses. This can be avoided with the help of preventive maintenance
 7. **Robustness and reliability of a machine**
 8. **Cost of implementing AI** on an edge device (infra-investment) and how effective it is going to be
 9. **Conditions needed to set up and run AI** on edge applications (Does it have to be battery-powered or not?)
- Note.** Many other drivers/parameters control the adoption of AI on edge. But these generally differ from one use case, or one industry, to another.

device, which is on the edge and has the capability to run ML algorithms or do predictive analytics in conjunction with the backend cloud.

There is cost involved but it is a tradeoff in terms of the outcomes and the ROI that you expect. If there is a proper balance of security, cost, and form factor, then the edge could be very secure.

According to Venkateswar Melachervu, there is no drastic change in hardware on the edge from what is implemented on the cloud; they are pretty much the same. Edge devices, particularly mobile devices, have to run one or two iterations. This generally causes the battery to drain very fast, which needs to be recharged frequently. This is a constraint in terms of battery life as the device has to run recursively and there is a lot of computational load.

Purpose-built AI on edge

The stack that defines edge could be divided into several parts: Hardware, AI-capable microcontrollers, AI cores (CNNs, BNNs, accelerators, and other neural network binary codes), whether it is open-source or from Google. And then there are software tools that

could cross-compile high-level C++ or Java code onto these computing cores, and some reference designs and architectures and custom services. There are a lot of combinational architectures that Google and Azure are coming out with (both edge and hybrid).

Today, data storage is not a serious problem as gigabytes and terabytes of space is available pretty much on every device. The battery life is also gradually improving in terms of longevity. But the biggest challenge would be to have optimised or purpose-built algorithms for edge devices, IoT devices, gateways, etc.

This is where innovations are taking place. On the hardware level as an example, Google's AI-edge stack has an edge TPU, called CoralEdge, and an IoT edge software pack that brings in the entire Tensor framework algorithmic pieces as lightweight into mobile devices.

The iPhone 12 has A14 chip that is purpose-built for AI and ML. Called Deep Fusion, it is hybrid and has 8-core neural engines, 6x faster matrix multiplication, one trillion hours per second on CPU, and 40 per

cent more power efficiency, making the compute cycle and instruction execution efficient. NVIDIA is also not far behind. Facebook has PyTorch mobile. In terms of difference, each is optimised for battery, storage, and other device needs.

The currently released GPUs are basically vector processing units built into the SoC itself. It is like integrating GPU inside the CPU, and giving a combined solution on the edge, which can do all of those calculations really fast.

Solving the latency issue

For the data, which is being produced at millisecond-level or even second-level, you need some kind of analytics on the edge so that you can prioritise and make immediate decisions, and choose to store aggregated data.

There are two dimensions to latency: propagation delay (the time taken for data to go and come back, which is decided by your IP routes and routing cables) and the time taken to run compute-intensive AI and ML algorithms like CNNs and neural networks. So, if you are running resources in a high-computing data centre with mainframes, then your time would be relatively less compared to a normal phone, which has a limited MIPS capability.

Companies like NVIDIA, Google, and even Apple have purpose-built AI-chips for running heavy computational loads with respect to AI and ML. Google's TFUs and TPUs run at a much faster rate on par with mainframes and big machines. These tools are coming to the edge today as there is a need and value for businesses and people. It is essential to have hardware with good network and compute speed along with the technical stack such as TensorFlow.

Whenever designing or developing an AI or ML system, you should take into consideration the required response speed along with the